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The Five Gs for Teaching Statistics: Greek, Graphs, Grammar, Gadgets, and Games

Andrew Dean Ho, *Harvard Graduate School of Education*

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Teachers of quantitative research methods face multiple challenges in advancing quantitative proficiency, including varying levels of comprehension, engagement, and belonging among incoming students. Consistent with consensus in the learning sciences, I introduce a mnemonic that can remind teachers and students to engage around multiple representations of statistical concepts. “The Five Gs” stands for Greek, Graphs, Grammar, Gadgets, and Games. The first three Gs, Greek, Graphs, and Grammar, refer to mathematical, graphical, and written representations, respectively. The last two Gs refer to Gadgets and Games, which I use to refer to tangible objects and friendly competitions, respectively. I locate the Five Gs in relevant learning sciences literature and demonstrate their application through teaching correlation coefficients. I conclude with guidance for judicious implementation in quantitative methods instruction.

Keywords: teaching, statistics, psychometrics, education, learning science

Author bio: Andrew Dean Ho is the Charles William Eliot Professor of Education at the Harvard Graduate School of Education. He is a psychometrician whose research aims to improve the design, use, and interpretation of test scores in educational policy and practice.

IX
When the blackbird flew out of sight,
It marked the edge
Of one of many circles.

-- Wallace Stevens (1954). “Thirteen Ways of Looking at a Blackbird."

I taught creative writing before graduate school. One of my favorite activities was inspired by the 1954 Wallace Stevens poem, “Thirteen Ways of Looking at a Blackbird.” In 13 brief stanzas, Stevens evokes different senses from his observations of a single subject, the blackbird. Each stanza makes the blackbird seem ominous, then ruminative, then vivid, then serene, and more. When teachers assign students to write their own “thirteen ways” poem about a subject that interests them, the hope is that students take different perspectives, ideally perspectives that they may not have held before. This can lead to deeper appreciation and understanding of the subject, for students and for those reading their poems.

I have similar goals for concepts that I teach in my statistics and psychometrics courses. From simpler concepts like central tendency and variance to more complex frameworks like causal inference and Generalizability Theory, I want my students to appreciate, if not 13, at least multiple representations of a statistical model and technique. When students learn to use these methods, I hope they can similarly engage audiences with the implications of their findings using multiple representations.

A challenge to multimodal student appreciation for statistics and psychometrics is the historical and current use of these quantitative methods for selection, exclusion, and discrimination (American Psychological Association, 2021). Many students arrive in my

courses believing statements like, “I am not a math person,” or “I am not good with numbers.” Statistics and quantitative methods remain a “language of power” in social science research.[1] If students are not fluent in this language, they are susceptible to further exclusion by those who are fluent. Teaching educational research methods can disrupt and change these beliefs to promote inclusion in the academy and research professions, improving methodological pluralism, communication, and belonging among future researchers.

This paper introduces a “Five Gs” mnemonic that I argue can help teachers to address varying levels of student comprehension, engagement, and belonging. The mnemonic should remind teachers and students to teach and learn using multiple representations of statistical concepts. Respectively, these “Gs” encourage teachers to design interactions that:

1) Represent concepts mathematically, using **Greek** letters when appropriate, following statistical conventions.
2) Represent concepts **Graphically**, using figures aligned to mathematical notation, enabling visualization of descriptive or causal relationships.
3) Represent estimates **Grammatically**, that is, write a coherent sentence that interprets the magnitude of the estimate and its implications, in context.

[1] By “language of power,” I mean a learnable vocabulary and grammar that confers status and belonging to those who are fluent. This concept parallels Delpit’s (1995) emphasis on explicitly teaching the “codes” of the “culture of power” and Bourdieu’s (1986) “cultural capital.” I prefer “language” because it foregrounds learnability and positions statistical fluency as a goal of inclusive teaching.

4) Represent concepts tangibly, using physical, ideally "real-world" representations of concepts in the form of **Gadgets** or manipulatives.
5) Represent concepts in the form of a friendly challenge or competition, a **Game** where expertise matters while allowing students to be comfortably incorrect.

In many statistics classrooms, the order of the Five Gs reflects the rank order that teachers and students value and spend their time. Presentations emphasize Greek: statistical models, their parameters, and their estimates. Graphs and figures are common but secondary. And written "model language" that interprets the magnitudes of estimates and their relevance is rare. Physical manipulatives (gadgets) and statistically oriented games are a one-off special activity or written off as gimmicks. This paper aims to increase awareness of this imbalance and rebalance the scales toward multiple, integrated representations.

The 5 Gs: Greek, Graphs, Grammar, Gadgets, and Games

Early in my career teaching quantitative methods, I would tell students that I valued multiple representations, including mathematical, graphical, and written representations. However, I would often forget to practice what I preached, tending towards didactic lectures that were too mathematical and monotonous. In particular, I often forgot the "Grammar" for estimated parameters, by which I mean explicit written interpretation of the magnitude of estimated quantities, in context. As my colleague Carrie Conaway (2019) states, the goal is to write "a sentence with a number in it;" or "measurement must be qualitative, then quantitative, then qualitative again" (Ho, 2024, p. 137). Students often requested this "model language" to answer the "so what" question in their minds. They also

felt it helped them to transfer their learning to required assignments and, ultimately, their communication goals for their own research.

In February 2020, a student in my measurement class, Eli Godwin, suggested that I adapt my "mathematical, graphical, written" representation goals into a "Greek, Graphs, and Grammar" mnemonic (personal communication). This helped me to remember to engage around these representations more often than I had. A "missing G" began to feel more like an urgent, missed opportunity to engage students toward learning.

Eli then quipped about my often-impromptu tendency to incorporate tangible objects into my presentations. When it seemed that I or my students were flagging, I would, for example, reference blueberries that a student happened to bring in as a snack as a metaphor for sampling. Or, I would bring out a hand-stitched normal distribution "plushie" that I kept in my bag. He suggested I add "gadgets" or "gizmos" for a "4 Gs" mnemonic. This addition spurred both me and my students toward incorporation of tangible objects into presentations. I used a large slinky to illustrate nonlinear transformations. And I created a play-dough mountain to give students a tangible sense of how maximum likelihood estimation would work in a two-parameter likelihood surface. These demonstrations always seemed to increase engagement.

The last G, for "Games," emerged after I developed asynchronous videos for my courses during the COVID-19 pandemic. When in-person instruction returned, I "flipped" my classroom to require students to watch asynchronous videos and then come to class for opportunities for more "active learning" (Freeman et al., 2014, p. 8410). I received feedback that the sum of asynchronous videos and synchronous sessions was beginning to

exceed the number of credit hours. In response, I made one of my twice-weekly, in-person sessions “optional and encouraged.” To incentivize the optional session, I included a “game” every week, with the chance of winning a prize, typically a content-related book. I believe that the only slight decline in attendance between the two sessions reveals some student interest in these additional learning (and/or winning) opportunities. I added Games to complete the Five Gs mnemonic that I present in this paper.

A brief review of literature related to the Five Gs

Recent consensus on how people learn emphasizes that effective instruction enables active learning with guidance (National Academies of Sciences, Engineering, and Medicine, 2018). It can combine elements of both “direct instruction” (explicit teaching of concepts and procedures) and “discovery learning” (opportunities for students to actively explore; Lee & Anderson, 2013). The Five Gs framework is compatible with this integrated approach. The first three Gs (Greek, Graphs, and Grammar) are presentable in texts, slides, and lectures (direct instruction), or through more discovery-based activities where students generate their own representations. The challenge in any instructional presentation is to carefully integrate representations while attending to the constraints of the topic and realistic limits on learners’ working memory (Ainsworth, 2006; Chandler & Sweller, 1991).

Gadgets can also be consistent with approaches to direct instruction and discovery learning. Calls for exploring with interactive, tangible “gadgets” date back to Maria Montessori (1912) and Froebel’s Gifts (Wiggin & Smith, 1895). More recent studies find positive effects of manipulatives in more direct math instruction (Carbonneau et al., 2013;

Donovan & Fyfe, 2022). Constructionist philosophies suggest that such gadgets should be student-led or co-created, not just teacher-created and teacher-controlled;[2] however, rigorous evidence of, for example, the proliferation of makerspaces on outcomes is scant (Soomro et al., 2023).

The games I use in my teaching are deliberately low-stakes and formative, and most have some element of luck.[3] This contrasts with and could complement other notions of "games." For example, "gamification" uses reward systems to incentivize individual effort (see Dicheva et al., 2015). And medium-stakes collaborative assessments, including the "immediate feedback assessment technique" (IF-AT; Epstein et al., 2002), two-stage exams (Balch, 1992; Wieman et al., 2014), and collaborative exams (Jang et al., 2017) have shown promise for improving learning through peer interaction. My approach to games is similarly compatible with both direct instruction and discovery learning. Instructors can provide more or less guidance about winning strategies depending on learning goals.

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[2] Over the past 3 years, I have allowed (but not required) students to work with me to create their own "gadgets" (or games) as a final project option. One student has taken me up on the opportunity thus far. For Gadgets (or Games) to demonstrate a student's statistical proficiency, I recommend requiring students to develop and accompanying written and/or oral presentation that explains how instructors and learners can use them to teach and learn.

[3] For example, in a Classical Test Theory ($X = T + E$) game in my measurement class, I pass out a coin and a six-sided die to every student and have each of them "roll" a true score T that can take a value of 1-6. I then have them "roll" subsequent error terms E_1, E_2, … while flipping a coin that makes the errors positive or negative, so errors range from -6 to +6. Students with different true scores gain an intuitive sense of how "unfair" error variance can be, while appreciating how averaging over errors increases reliability and precision.

Because theoretical die rolls are a discrete uniform distribution, the true score variance is known and equal to $35/12$. The error variance is nearly discrete uniform from -6 to 6 but excludes 0 by construction and is thus equal to $182/12$. Dividing this by the number of rolls then gives a theoretical reliability of one roll of .16 and of 4 rolls of .43.

This game is entirely luck. I tell students that I may give the prize to the highest observed score or the lowest, or to the highest average error or the lowest, or to the student with the average error closest to 0. Most of my other games involve some amount of skill, like the one I describe in my illustrative example, but they all have intentionally low stakes.

If a student wins a game twice in a semester, the prize goes to their assignment partner.

My emphasis on low stakes derives in part from the history of exclusion in statistics and psychometrics that I mention above. Although I have higher stakes assignments, oral exams (Ho, 2025), and final projects, I also want to ensure that in-class sessions remain a community where mistakes and misinterpretations are not penalized. In these formative spaces, I hope mistakes are acceptable if not outright welcomed steps on a learning journey. Consistent with the literature on growth mindset (Burnette et al., 2023) and Dweck's (2006) emphasis on the "power of yet," I begin the semester by introducing what I call the learner's mantra: "I am not afraid to be wrong; if I am wrong, it just means I haven't learned it yet." By retaining low-stakes games with an element of luck throughout the semester, I hope I give learners many opportunities to repeat and internalize this mindset, while I model it with my own (often incorrect) guesses in these games, myself.

An example of the 5 Gs: Teaching the meaning and use of correlation coefficients

I teach correlations at the beginning of the semester in a brief review intended to ensure that the second-semester educational researchers (who have taken different first-semester courses) have a common foundation for future learning. I follow my colleagues Judy Singer and John Willett's (1990) longstanding advice to teach using real data examples. I ask students to consider claims like, "test scores only measure income," and "your zip code should not determine your educational opportunities." I ask whether they believe the relationship between socioeconomic status and educational test scores is weak, moderate, or strong in the United States.

To make the example concrete, I introduce them to an open-access dataset I helped to create: the Stanford Education Data Archive (SEDA, Reardon et al., 2024). I use a version

of SEDA that includes district-level average test scores from grade 3-8 state testing programs, from 2009 to 2019, as well as a district-level socioeconomic status indicator (Fahle et al., 2024). I ask students to guess the relationship between district-level test scores and district-level socioeconomic status on a scale of 0 to 1. This becomes a game of "Guess the Correlation." I ask students to write down their guess before proceeding through the 5 Gs in turn.

The First G: Correlation in "Greek"

I walk students through the various formulas for a correlation. The formula for calculating correlations from raw data is counterintuitive, so it is important to establish the equivalence to standardized covariance, a scaled dot product of standardized scores, and a standardized regression coefficient. I show these relationships here using standard notation. Instructors can use animated slides to show how these equations morph into each other with known equivalencies. For additional historical, mathematical, and geometric perspectives on the correlation coefficient (that also alludes to the Wallace Stevens poem), see Rodgers and Nicewander (1988).

$$r_{xy} = \frac{\sum\left((x_i - \bar{x})(y_i - \bar{y})\right)}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} = \frac{\sum\left((x_i - \bar{x})(y_i - \bar{y})\right)}{(n-1)s_x s_y} = \frac{Cov(x,y)}{s_x s_y} = \frac{\sum\left(z_x z_y\right)}{n-1} = b_{y|x}\left(\frac{s_x}{s_y}\right)$$

The Second G: Correlation in "Graphs"

I then provide the graph in Figure 1, namely a scatterplot of district average test scores on the standardized socioeconomic status indicator from SEDA. Each observation is one of the 11,926 U.S. school districts with nonmissing estimates, and the size of the dot corresponds to the number of test scores contributing to the estimate. I gloss over details

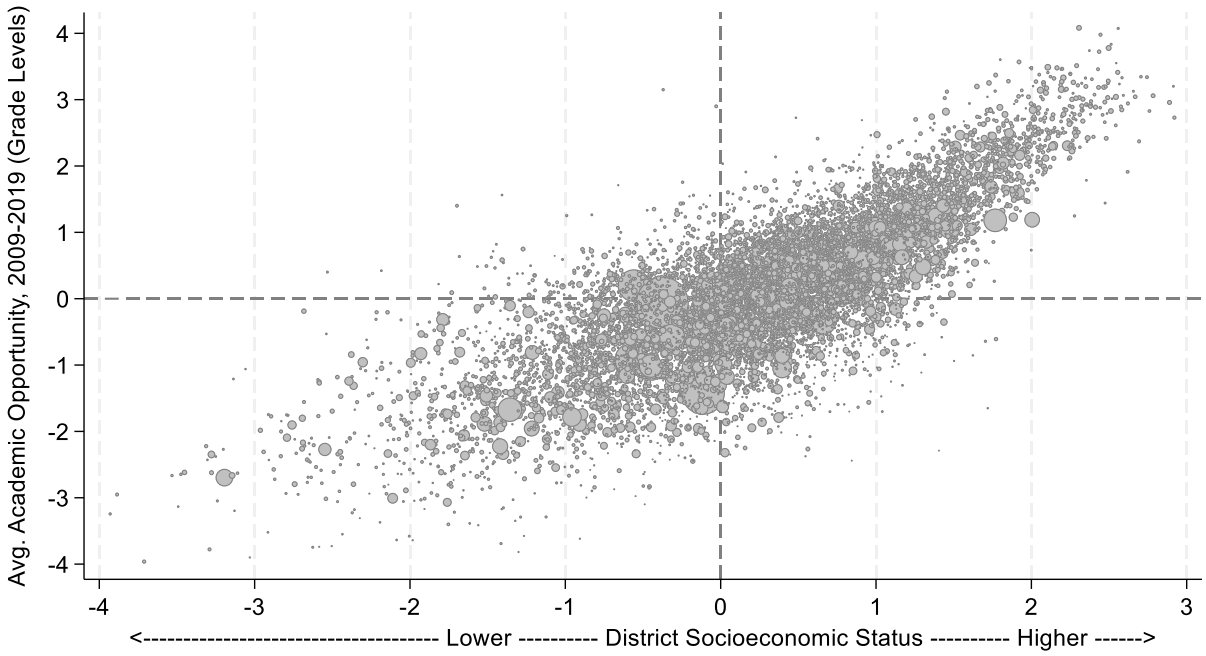
related to the aggregate nature of these data.[4] The goal is to help students visualize the relationships represented in the equations above.

I create axes shown by the dotted lines in Figure 1, at the means of both variables. I help students visualize how observations in each "quadrant," defined by these axes, contribute positively or negatively to the correlation. Observations in the upper right and lower left "quadrants" contribute positively to the correlation, and observations in the upper left and lower right "quadrants" contribute negatively, per Equation 1. I then show an interactive graph with a slider that controls a simulated scatterplot with correlations between -1 and 1, so students can appreciate what degrees of correlation look like between the extremes. Given this, I ask students to discuss their originally guessed correlations with their neighbors and modify their guess as they wish.

Figure 1

Scatterplot of District Average Test Scores and Socioeconomic Status

[4] I emphasize to students that these are district-level correlations that are different (substantially higher) than correlations between student test scores and student socioeconomic status. I also explain that the SEDA estimates I use for this exercise are Empirical Bayes (imprecision-shrunken) estimates per Fahle et al. (2024). SEDA also includes ordinary least squares estimates with standard errors suitable for precision-adjustment of correlations. I explain to students that we ignore these complexities for now and that these issues motivate the multilevel modeling and measurement error units later in the semester.



Note. Data from 11,926 United States school districts from 2009 to 2019. Each dot represents one district, with dot size corresponding to the number of test scores contributing to the estimate. Source: Stanford Education Data Archive (edopportunity.org).

The Third G: Correlation in "Grammar"

I reveal that the (naïve, district-level) correlation coefficient is 0.78. Following the goal to "write a sentence with a number in it," I explain that simply saying "the correlation is 0.78" is not sufficient. What does this magnitude mean? Students sometimes return to the slider animation to set mental anchor points between 0 and 1. They state that the relationship is 78% of the distance between no relationship and a perfect linear relationship. Some also use their regression knowledge to square the correlation and express the resulting R^2 value in terms of a percentage of variance. I return to Equation 1 to remind students that the correlation coefficient is a regression coefficient for standardized

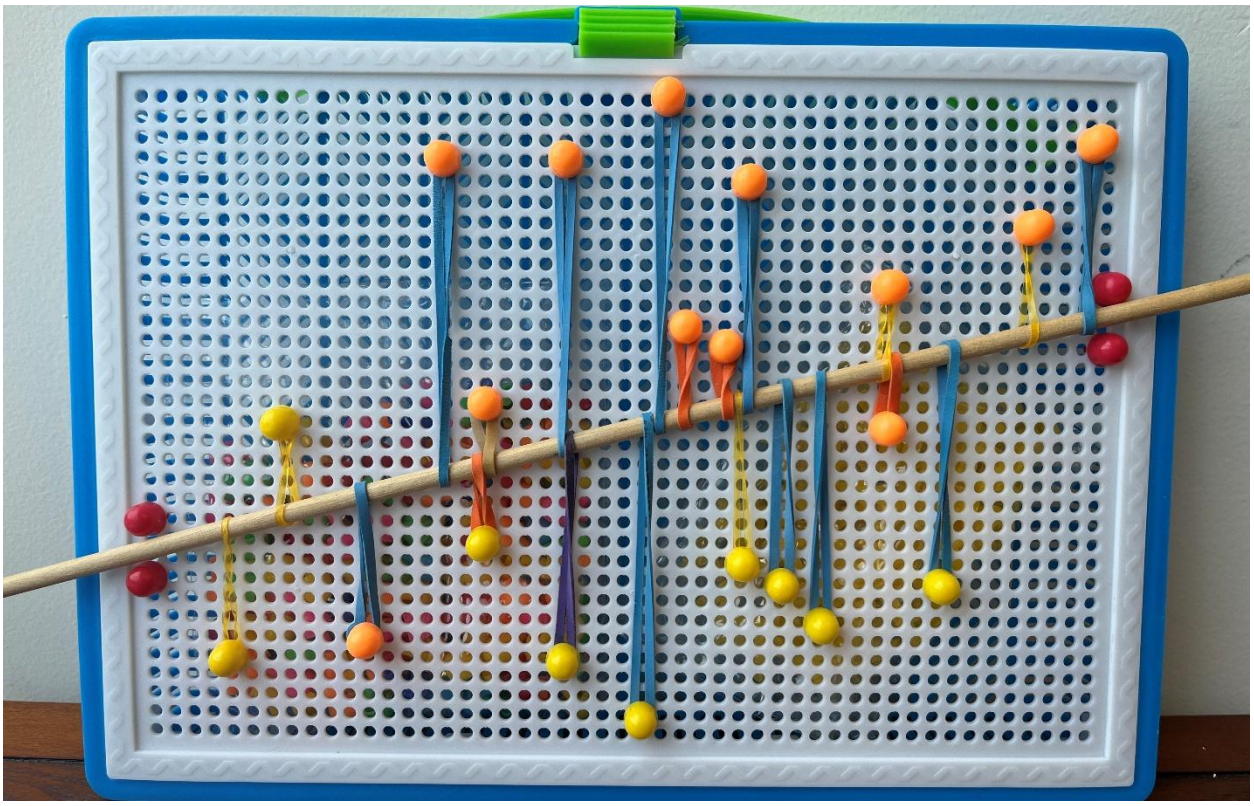
variables. This leads to the somewhat lengthy but accurate sentence, “I predict two districts that differ by 1 standard deviation unit in socioeconomic status will differ by 0.78 standard deviation units in test scores, on average.”

The Fourth G: Correlation in “Gadgets”

Earlier in my accelerated review of introductory material, I use a “rubber bands and broomsticks” analogy to explain ordinary least squares (OLS) regression. I use a real broomstick to help students visualize that the regression slope minimizes the “vertical (squared) stretch” of rubber bands connected from each observation to the broomstick. Sometimes I have students represent these observations with rubber bands pulling on the broomstick from different directions to show how outlying observations can change the slope of the regression line. I return to this “gadget” here, either as a broomstick and large rubber bands, or with a dowel attached to smaller rubber bands in a pegboard. Figure 2 illustrates what this manipulative can look like in action. The key concept to bridge from regression to correlation is that the correlation is the slope of the regression line after variables are standardized. Labeling units in both the unstandardized and standardized metric on the pegboard can illustrate this.

Figure 2

A “Rubber Bands and Broomsticks” Model of Regression and Correlation



Note. The pegboard represents observations, the wooden dowel represents the regression line, and rubber bands represent residuals. The model provides intuition for ordinary least squares regression coefficients. Instructors can then connect this slope to a correlation when variables are standardized.

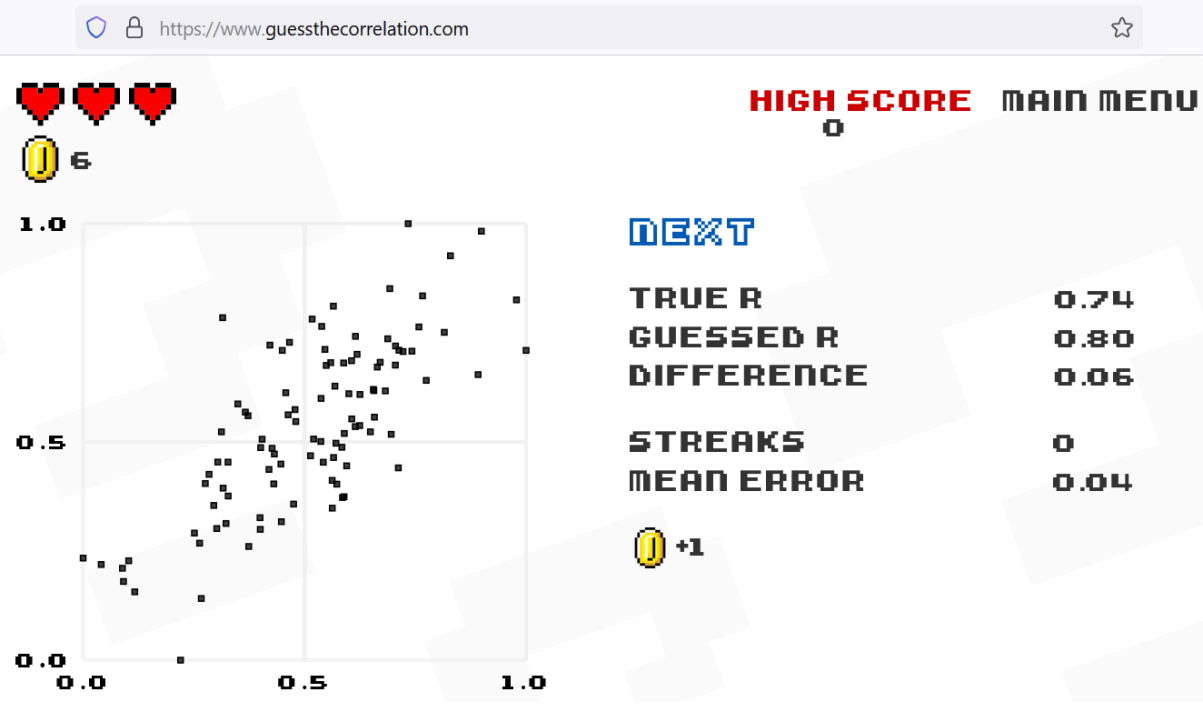
The Fifth G: Correlation in "Games"

I close this set of interactions with a "Guess the Correlation" game. Students try to guess the correlation in a randomly generated scatterplot. I give them credit for guessing closer than my own (often faulty) guesses. I use the website guessthecorrelation.com, which has a charmingly retro interface (Figure 3) that features 8-bit graphics and arcade-style scoring. I show a scatterplot and encourage students to discuss their guesses and

their reasoning with their neighbors. This typically generates lively discussion. I have them submit their guess via a google form for some monitoring, as students sometimes misremember their own guesses. I then enter my guess, submit my answer, reveal the actual answer, and give a “point” to any student who had a guess as close as mine or better (in absolute distance) from the actual correlation. I remind students about the “learner’s mantra” and show by example how I can be wrong, even after years of practice. I give folks a warmup round or two. I gift a content-relevant book to students who have the most points after 3 or so rounds, including ties.

Figure 3

The “Guess the Correlation” Game Interface



Note. Screenshot from guessthecorrelation.com showing the interface students use to practice estimating correlation coefficients from randomly generated scatterplots.

Cautions and Caveats in Using the Five Gs

All teaching improvements require time, effort, and resources to be successful. Literature on the challenges of multimodal representations (Ainsworth, 2006; Chandler & Sweller, 1991) suggests that uncritical, incoherent, or rushed application of the Five Gs would be worse than a business-as-usual counterfactual. Instructors will need to cut back on some slides, topics, and goals to make room for more Gs. To do more, we must do less.

Like any first attempt at teaching, instructors must risk a game or gadget that is awkward or confusing before they learn how to motivate it clearly and organize it confidently. Not all efforts may ultimately be worthwhile. As an example, decades ago, my colleague Judy Singer developed an elaborate 3D model (using foam balls, pegboards and dowels) of a regression of graduate school ranking scores on two variables: doctoral graduate counts and average GRE scores. The model can help students to visualize a multiple regression function as a plane in 3D space. Rotating the model to visualize 2D projections can also help students to see how regression coefficients can change when we add additional confounded variables.

However, introducing this "gadget" was time consuming. I had to introduce Judy's historical dataset, regression model coefficients, and all associated context. The model is also large enough that it is not easily passed around a class of more than 50 people. I ultimately decided to use easier interactive online scatterplots (graphs, not gadgets) while making Judy's model available for out-of-class viewing and hands-on exploration.

Student sentiment can also challenge transitions to discovery and active learning, or to Games and Gadgets from "traditional" mathematical presentations. Deslauriers et al.

(2019) find that students in active classrooms believe they are learning less even when they are actually learning more. Students might feel as though they have a preferred type of representation and may feel disengaged or unmotivated by "other Gs." I find it helpful to emphasize that multimodal representation is an explicit learning goal of the course. At least, the first 3 Gs are nonnegotiable: I remind students that they must not only estimate but communicate quantities in both scholarship and practice.

For the last 2 Gs, teachers should always be prepared to explain what distinguishes a gadget or a game from a gimmick. To me, this requires that the gadget or game emphasizes if not reveals one or more explicit conceptual principles that I can communicate clearly. For example, in Figure 2, the OLS regression line is the slope of the wooden dowel, the correlation is the slope when variables are standardized, and these slopes minimize the sum of squared residuals represented by the cumulative stretch of the rubber bands. If the gadget or game is successful, the student should be able to explain how it represents or illustrates these principles. And teachers should consider why engagement around this principle is sufficiently central to learning goals to be worth the additional time that preparation and presentation of gadgets and games often require.

Still, as I've argued, the development and deployment of these two often overlooked representations can be engaging, memorable, and effective. At their best, they challenge and encourage instructors and students alike to see statistical concepts with new eyes. Beyond the example of correlations above, I have developed decent 5 Gs lessons for a variety of topics, including logistic regression, discrete-time event history (survival) analysis, regression discontinuity, principal components analysis, generalizability theory,

and item response theory. I am still struggling to develop effective 5 Gs representations for other topics, including instrumental variables, intraclass correlations, and Mundlak models. Coincidentally, these are techniques that I learned later in my own studies and for which I still may be building my own intuition. My hope is that the continuing search for such representations is also challenging me to improve my own understanding and, thereby, my teaching.

Resources and repositories can help. Collections like Gelman and Nolan's Teaching Statistics: A Bag of Tricks (2017), the CAUSEweb.org repository, activities from the Journal of Statistics Education, and simulation-based materials from rossmanchance.com include many resources suitable for adaptation to “Five Gs” goals. It is my hope that this journal, too, can inspire development, sharing, and implementation of worthy multimodal representations that advance the teaching of educational research methods.

Conclusions and Generalizations

Teaching with the Five Gs, like the “13 Ways of Looking at a Blackbird” activity, helps students see from complementary expert perspectives while encouraging them to develop and communicate their own expertise multimodally. We encourage students to engage beyond Greek, to Graphs that make relationships visible, Grammar that grounds numbers in meaning, Gadgets that make abstract concepts tangible, and Games that make exploration safe. I believe this creates more entry points for learners to access and wield essential tools of inquiry, an important act of inclusion in an historically exclusive discipline.

While I have illustrated the Five Gs in quantitative methods instruction, the underlying principles may have parallels when teaching other educational research methods. For example, qualitative methods instructors might consider analogous multimodal representations: teaching qualitative coding through visual mapping of themes, written memos that interpret patterns, tangible artifacts like concept cards that students physically arrange and rearrange, and collaborative coding "games" where disagreements become learning opportunities. The specific representations may differ, but instructor commitment to multiple entry points and multimodal communication may remain useful.

My development and ongoing search for the Five Gs across statistical topics also reflects values I hold in my teaching that may be shared regardless of which methods we might teach. First, students can be our best pedagogical collaborators. The Five Gs would not be so many nor so named without the responsiveness and creativity of my students. Second, effective teaching requires iteration and refinement over time. The mnemonic and each of its instantiations across topics have evolved gradually and not always smoothly: two steps forward, one step back.

Third, pedagogical innovation requires vulnerability. I risk awkward demonstrations, losing in games, and admitting when a carefully crafted gadget confuses more than it clarifies. This vulnerability, I believe, is itself pedagogical. It models for students that expertise includes comfort with uncertainty and willingness to revise approaches when evidence suggests better alternatives. In these ways, I hope the Five Gs both reflects and represents values that generalize beyond teaching quantitative methods. It is less about a

specific mnemonic and more about systematic attention to whether our pedagogical choices align with our goals for student learning, engagement, and belonging.

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